School of Future Tech

Case Study Report

on

FleetCare: Vehicle Fleet Monitoring System

by

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1. Introduction to the Case Study

Fleet management has become an essential requirement for logistics companies, transportation businesses, and organizations that operate multiple vehicles. Managing a large number of vehicles manually can be difficult and inefficient. Important tasks such as tracking vehicle mileage, monitoring fuel consumption, scheduling maintenance, and identifying service alerts require an organized digital system.

This case study focuses on the design and development of **FleetCare: a Vehicle Fleet Monitoring System**. The system is implemented as a responsive web dashboard that allows users to monitor multiple vehicles, view maintenance alerts, track fuel consumption, and analyze vehicle performance.

The FleetCare platform demonstrates how modern web technologies such as HTML, CSS, and JavaScript can be used to build an interactive dashboard interface for managing fleet operations. The system presents important fleet information through structured layouts, tables, and dashboard cards to provide a clear overview of vehicle data.

This case study highlights the importance of organized interface design, responsive web layouts, and dynamic data handling in developing practical fleet management solutions.

2. Problem Statement / Case Background (Abstract)

Background

Organizations that manage a fleet of vehicles often face challenges such as tracking vehicle usage, maintaining service schedules, monitoring fuel efficiency, and detecting maintenance issues. Without a centralized system, these tasks become time-consuming and prone to errors.

Modern fleet management systems provide digital dashboards that allow managers to monitor vehicle information in real time. Such systems help organizations improve operational efficiency, reduce maintenance costs, and ensure vehicle safety.

Abstract

This case study presents the design and implementation of **FleetCare**, a web-based vehicle fleet monitoring system. The system provides a centralized dashboard to monitor multiple vehicles, track mileage, record fuel consumption, and generate maintenance alerts.

The platform consists of multiple modules including a fleet overview dashboard, vehicle monitoring page, maintenance tracker, fuel monitoring system, and alerts/reporting section. The

application uses responsive layouts, dashboard cards, structured tables, and JavaScript-based alert logic to present data effectively.

By integrating a clean user interface with dynamic data updates, the system demonstrates how web technologies can be used to build a practical solution for managing fleet operations.

3. Case Study Design

The FleetCare system is designed as a **multi-page web dashboard** that organizes fleet management features into separate modules.

The system architecture includes the following components:

Fleet Overview Dashboard

The homepage provides a summary of fleet statistics including:

- Total number of vehicles
- Active maintenance alerts
- Fuel efficiency statistics
- Vehicles currently in service

This overview helps users quickly understand the overall status of the fleet.

Vehicle Monitoring Module

This module displays individual vehicle information using tables and cards, including:

- Vehicle identification
- Mileage details
- Service status indicators

Vehicles that require maintenance are highlighted to draw user attention.

Maintenance Tracker

The maintenance module helps track upcoming service schedules and maintenance history. It displays service due dates and alerts for overdue maintenance.

Fuel Monitoring System

The fuel monitoring module records fuel consumption and calculates fuel efficiency values for each vehicle.

Alerts and Reports

This module displays warnings related to vehicle maintenance, mileage limits, and performance summaries.

User Interface Design

The website uses:

- Responsive layout for different screen sizes
- Dashboard cards for key statistics
- Structured tables for detailed information
- Navigation bar and footer for consistent user experience

4. Methods & Algorithms Technology Applied in the Problem Statement / Case Study

The FleetCare system uses several web development techniques to manage and display fleet data effectively.

Interface Design Methods

Dashboard Cards

Dashboard cards are used to display important fleet statistics in a compact and visually clear format.

Structured Data Tables

Tables are used to organize vehicle data, fuel logs, and maintenance schedules.

Responsive Layout

Responsive design ensures the website works across devices such as desktops, tablets, and smartphones.

JavaScript Logic

The system uses JavaScript to implement basic dynamic functionality such as:

Mileage Alert Logic

Vehicle mileage is checked against a predefined service threshold. If the mileage exceeds the limit, the system highlights the vehicle as requiring service.

Dynamic Data Display

JavaScript updates certain dashboard elements dynamically based on fleet data.

Navigation and Interaction

JavaScript also helps control interactive elements such as navigation behavior and dynamic content updates.

Technology Stack Used

Programming Languages

- HTML
- CSS
- JavaScript

Web Development Techniques

HTML
Used to structure the website layout and define page content.

CSS
Used to style the interface, create responsive layouts, and apply the corporate blue/gray theme.

JavaScript
Used for dynamic functionality including mileage alert logic and interactive UI behavior.

Development Environment

- VS Code or any modern code editor
- Web browser for testing and debugging

Hosting Platform

The project is deployed online using
Vercel.

5. Problem Statement / Case Study Implementation Details and Snapshots

The FleetCare project is implemented as a **five-page web application**, with each page representing a specific fleet management module.

Project File Structure

The project contains the following main files:

- index.html – Fleet overview dashboard
- vehicles.html – Vehicle monitoring page
- maintenance.html – Maintenance tracking page
- fuel.html – Fuel monitoring page
- reports.html – Alerts and reports page
- style.css – Website styling and layout
- app.js – JavaScript logic for alerts and interactions

Fleet Overview Dashboard

The homepage displays summary information about the fleet using dashboard cards.

Example information displayed:

- Total vehicles in the fleet
- Number of active alerts
- Average fuel efficiency
- Vehicles currently under service

Vehicle Dashboard

The vehicle dashboard displays detailed vehicle information using tables.

Data displayed includes:

- Vehicle name or ID
- Mileage information
- Service status indicator

Vehicles requiring maintenance are highlighted using alert colors.

Maintenance Tracker

This page shows upcoming maintenance schedules and overdue services. It allows users to easily identify vehicles that require servicing.

Fuel Monitoring Page

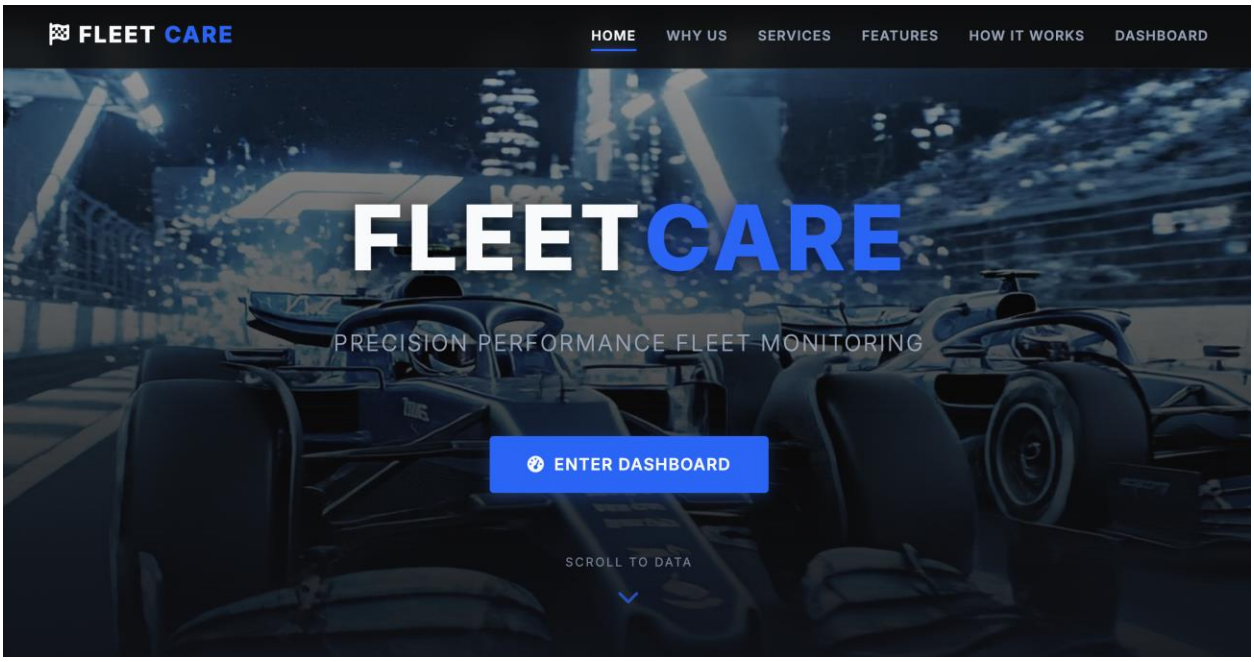
The fuel monitoring module displays fuel usage logs and efficiency calculations for vehicles.

Alerts and Reports

This module lists maintenance warnings and performance insights related to fleet operations.

Snapshots

- **Landing Page**


FLEET CARE
HOME
WHY US
SERVICES
FEATURES
HOW IT WORKS
DASHBOARD
FLEETCARE
PRECISION PERFORMANCE FLEET MONITORING
ENTER DASHBOARD
SCROLL TO DATA

Why Choose FleetCare
Real-Time Fleet Monitoring
Track your vehicles with live GPS mapping, telemetry data, and instant speed updates.
Automated Alerts
Stay ahead of maintenance with predictive alerts based on vehicle mileage and wear.
Fuel Efficiency Tracking
Analyze fuel consumption to dramatically reduce operating costs and optimize routing.
Performance Analytics
Generate detailed PDF reports highlighting the most and least efficient vehicles across the fleet.

Our Services

Fleet Monitoring

Get full visibility over complex logistics networks to ensure timely deliveries and compliance.

Maintenance Tracking

Ensure all trucks are in peak operational condition to avoid costly downtime and repairs.

Fuel Analytics

Drill down into fuel usages to find anomalies and drivers needing additional training.

System Capabilities

- Live Monitoring
- Mileage Tracking
- Service Alerts
- Fuel Analytics
- Performance Reports

How FleetCare Works

01

Add Fleet Vehicles

Register your trucks and drivers into the central data array to begin tracking.

02

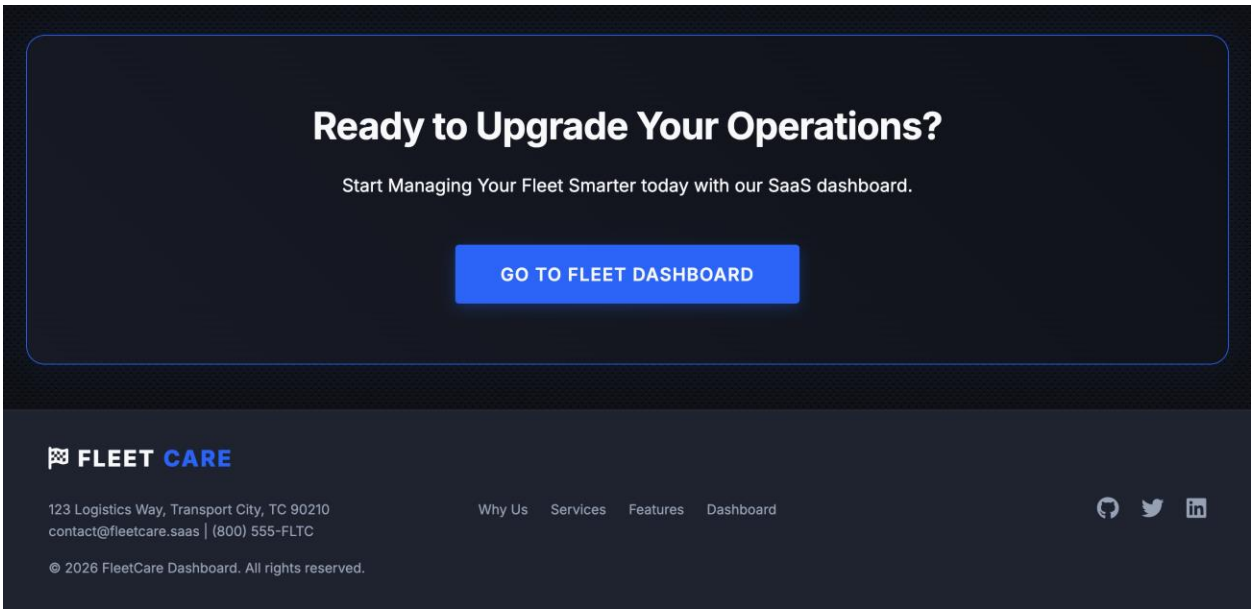
Monitor Data

Watch distance and fuel metrics update continuously across your SaaS interfaces.

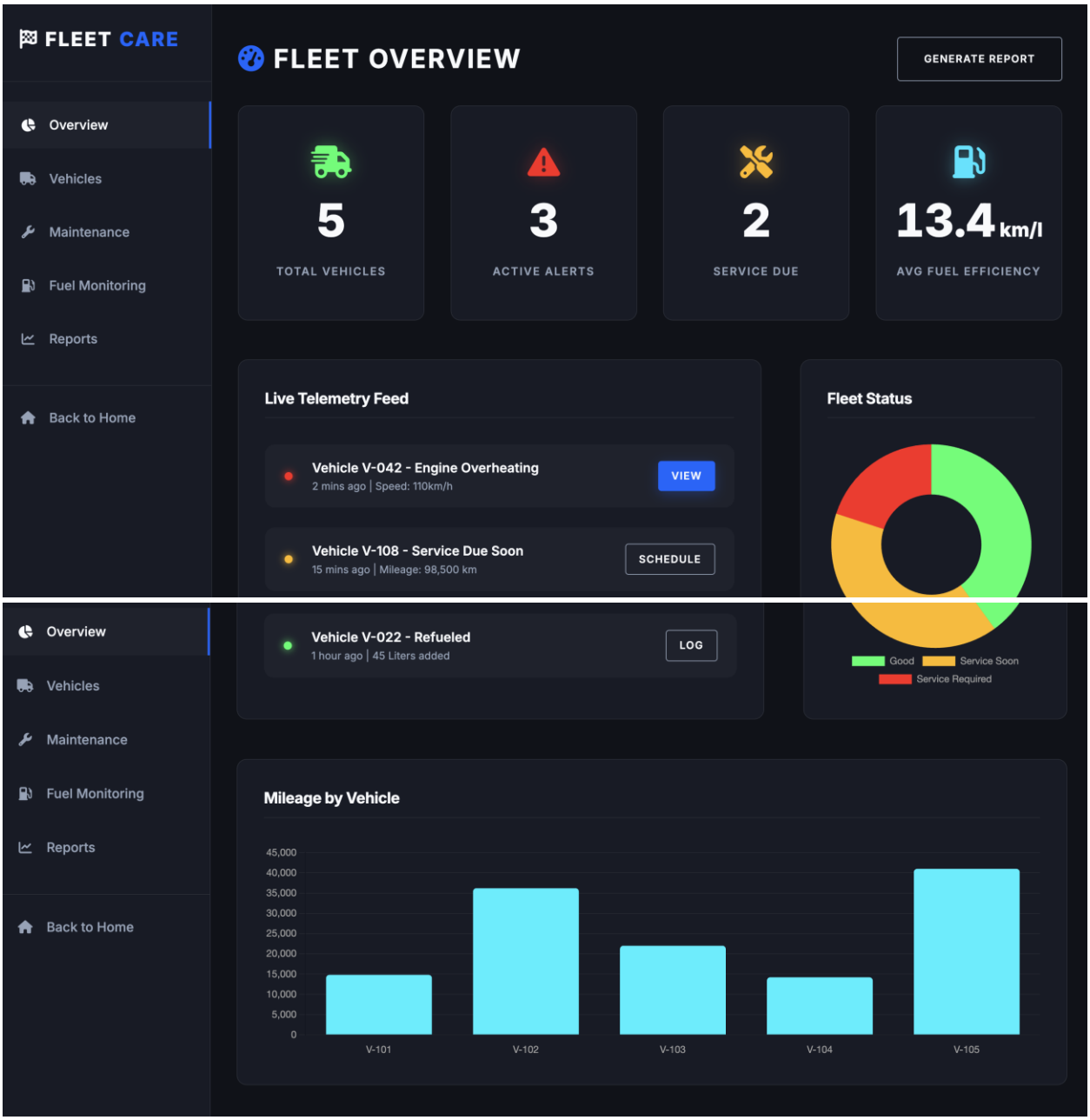
03

Receive Smart Alerts

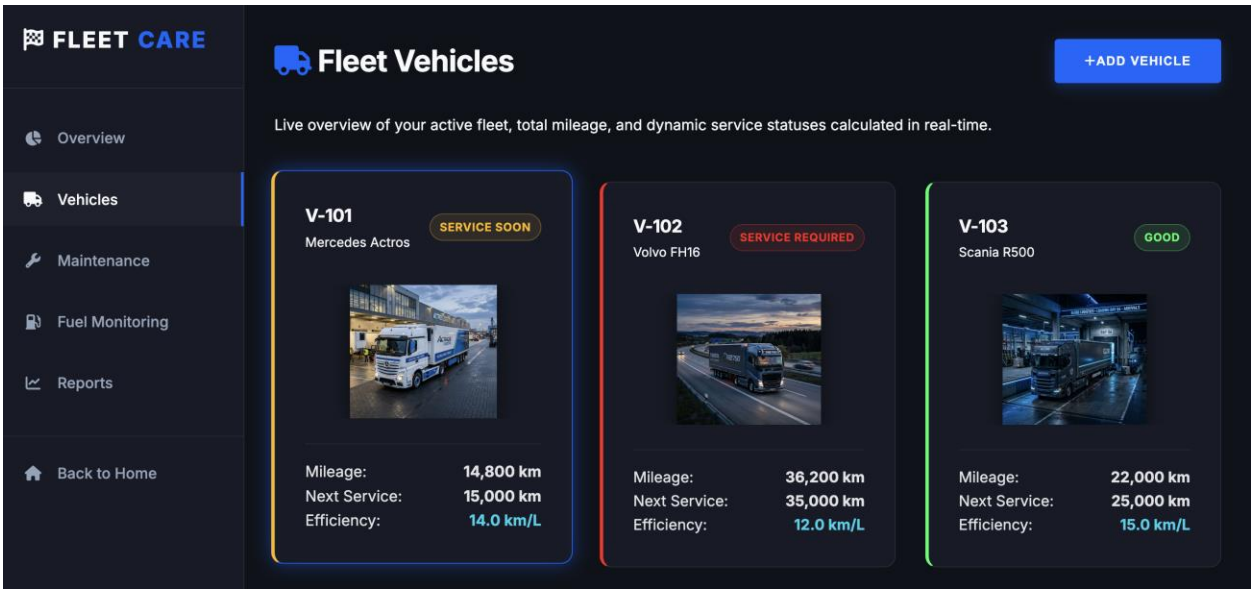
Let the algorithm automatically notify you when it's time for vehicle service.

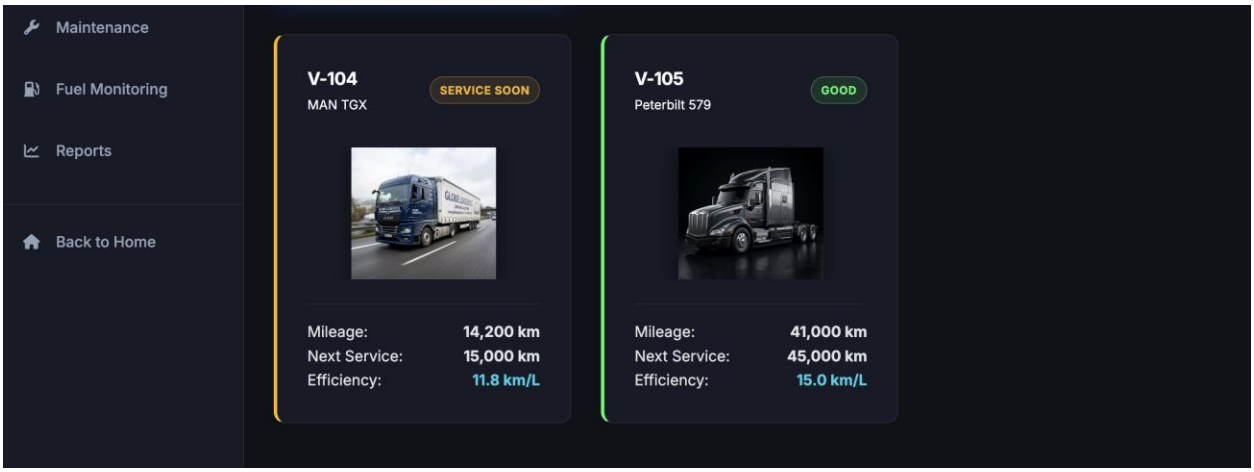


- **Fleet overview dashboard**

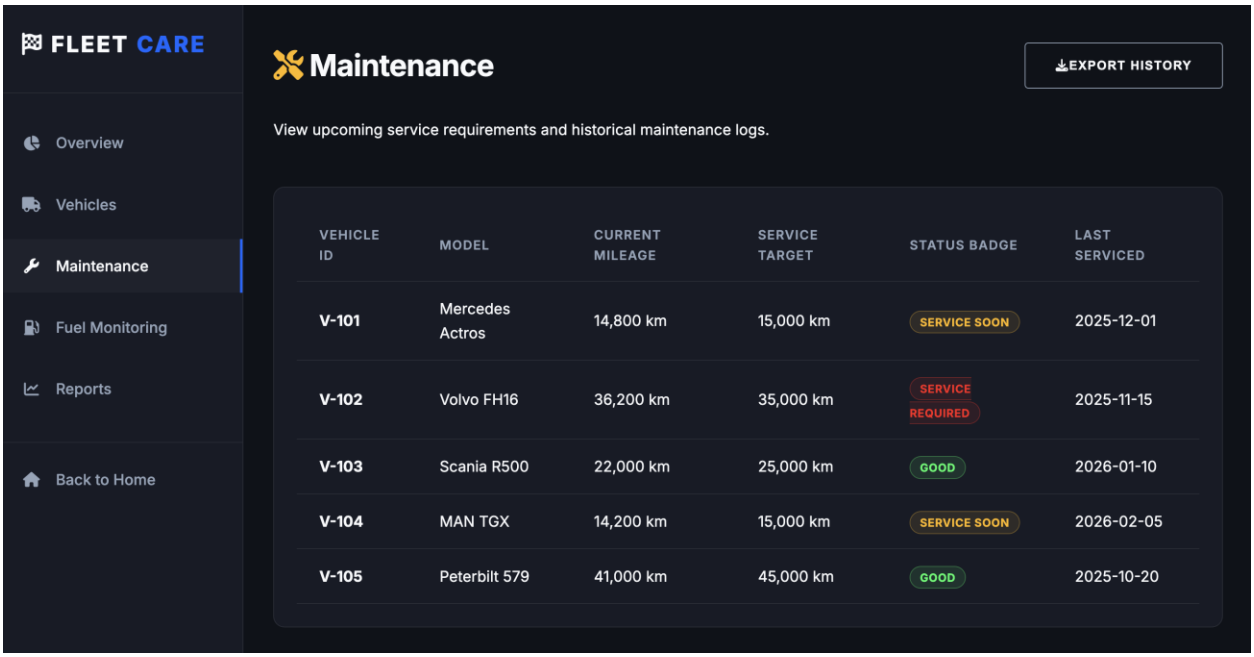


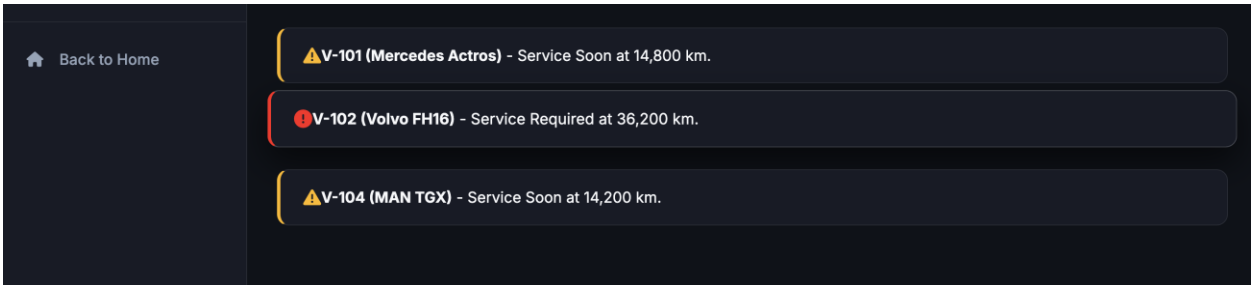
- **Vehicle monitoring page**



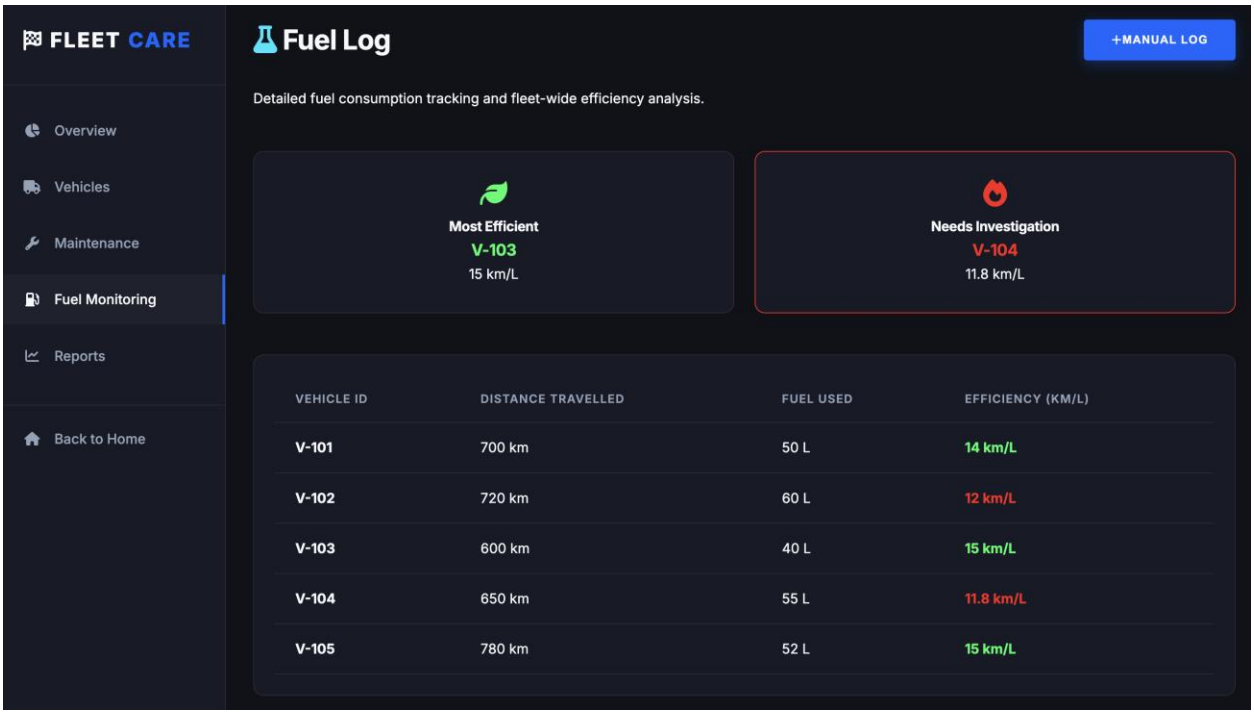


- **Maintenance tracker page**

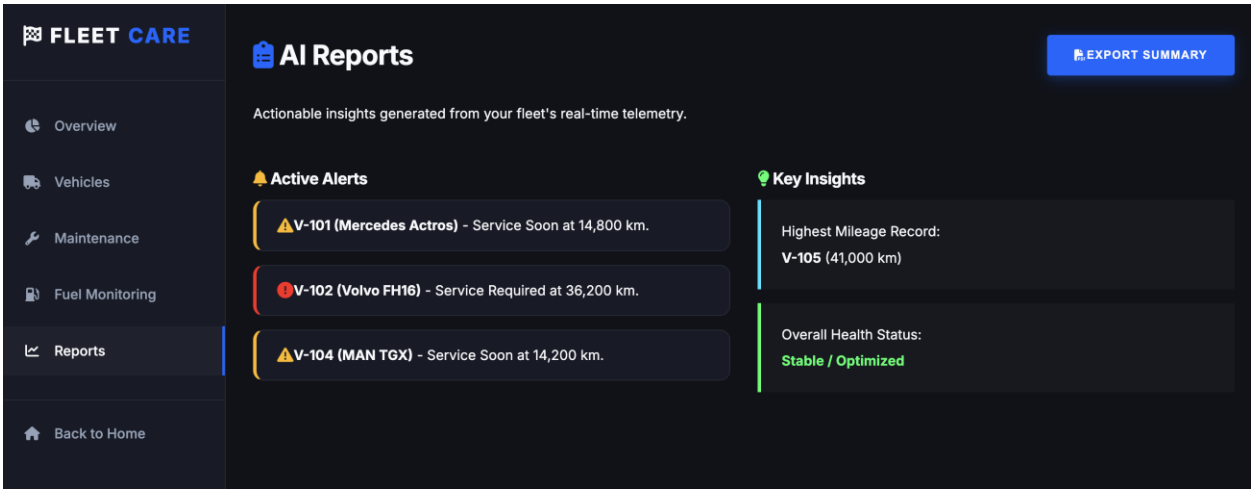




- **Fuel monitoring page**



- **Alerts and reports page**



6. Problem Statement / Case Study Results and Conclusion

Findings

The FleetCare system successfully demonstrates how a web-based dashboard can be used to manage vehicle fleet data. The structured interface allows users to quickly access important fleet information such as vehicle mileage, maintenance schedules, and fuel usage.

The use of dashboard cards, tables, and alert indicators improves readability and helps users identify important issues such as service due notifications.

Conclusion

This case study presents the design and development of **FleetCare: a Vehicle Fleet Monitoring System** using modern web development technologies.

The project integrates multiple modules including fleet overview, vehicle monitoring, maintenance tracking, fuel monitoring, and alert reporting. Through the use of responsive layouts and JavaScript-based logic, the system provides a functional and user-friendly dashboard for managing fleet operations.

The FleetCare system serves as a foundational prototype that could be expanded further by integrating real-time data, databases, GPS tracking, and advanced analytics for large-scale fleet management solutions.

7. References

- HTML documentation from the World Wide Web Consortium (W3C).
- CSS documentation for responsive web design techniques.
- JavaScript documentation for implementing dynamic webpage functionality.
- Web development tutorials and learning resources related to dashboard design and frontend development.
- Documentation of the deployment platform
  Vercel.